

ML Specification Report — playground-series-s6e2

Heart Disease Prediction • our from-scratch agent (GBDT pool + tree-search v3)

Figures grounded in the competition's `facts.json`, `config.yaml`, `eda_summary.json`, `STATUS.md`, and `experiments_tree_v3.json`.

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Overview

The task is to rank 270,000 patients by heart-disease probability (metric: ROC-AUC) on a synthetic dataset derived from UCI Heart Disease. We solve it with a from-scratch pipeline — a pool of gradient-boosted trees (LightGBM / XGBoost / CatBoost) refined by a four-stage ladder and a 60-node tree search that ends on a 37-member mega-blend. Our champion scores **CV OOF AUC 0.955529** (submission file rounds to 0.95553); the NVIDIA reproduce-agent, copying a public XGBoost/LightGBM/CatBoost k-fold kernel, reaches **0.9554** — effectively a **tie**, with our agent nudging fractionally ahead.

Why it matters. Cardiovascular disease is a leading cause of death worldwide; risk-ranking from routine clinical features supports triage and preventive screening — as decision support, not diagnosis.

Data

Purpose of Data. Predict whether a patient has heart disease (Presence vs Absence) — a binary-classification Playground Series episode (S6E2). **Data Format** is clean **tabular CSV**, all numeric columns. **Data Volume** is **630,000 training rows / 270,000 test rows**, with **13 raw predictive features** once `id` and the target are removed.

Data Quality is high: **no missing values** in either split, **0 duplicate rows**, and only modest train↔test mean shift (largest $\approx +0.83\%$ on `Number of vessels fluoro`). The class balance is mild — **44.834% positive** (`Heart Disease` = Presence), an **imbalance ratio of 1.23** — which keeps the metric (AUC) clean and makes class-weighting largely unnecessary (confirmed in training). The strongest EDA signals, all consistent with cardiology intuition, were:

- **Thallium** is the dominant raw predictor (point-biserial $r \approx +0.606$).
- **Chest pain type** ($r \approx +0.461$), **Exercise angina** ($r \approx +0.442$), **Max HR** ($r \approx -0.441$), **Number of vessels fluoro** ($r \approx +0.439$), and **ST depression** ($r \approx +0.431$) form the strong mid-tier.
- **Slope of ST** ($r \approx +0.415$) and **Sex** ($r \approx +0.342$) contribute moderately.
- **BP** is essentially uninformative ($r \approx -0.005$); **FBS over 120** ($r \approx +0.034$) and **Cholesterol** ($r \approx +0.083$) are weak — a notable contrast with textbook risk-factor expectations.

Annotation Guidelines. The label is `Heart Disease` $\in \{\text{Absence, Presence}\}$, mapped to $\{0, 1\}$ (1 = Presence). Submissions are **positive-class probabilities** scored by AUC — a ranking, not a hard 0/1 decision.

Feature Set. Feb's focused **33-feature** set (13 raw UCI columns + 20 engineered) is carried into the July run and used for the XGBoost/CatBoost pool members; the 20 engineered columns were, however, **rejected for the LGB member** by a same-folds prior check (see the Training section), so the champion LGB family runs on the **13 raw features** alone:

Group	Features
Raw UCI (13)	Age , Sex , Chest pain type , BP , Cholesterol , FBS over 120 , EKG results , Max HR , Exercise angina , ST depression , Slope of ST , Number of vessels fluoro , Thallium
Thresholds / bins	Age_decade , BP_high , Chol_high , Thallium_7
Heart-rate derived	HR_reserve , HR_pct_max
Interactions	Age_x_MaxHR , Age_x_STdep , Age_x_Vessels , Chol_x_Age , BP_x_Chol , BP_x_Age , STdep_x_Slope , STdep_x_MaxHR , Vessels_x_Thallium , Angina_x_STdep , ChestPain_x_Angina
Composite / ratios	Risk_score , Chol_per_Age , BP_per_Age

Splitting strategy. A single canonical **Stratified 5-Fold CV (seed = 42, shuffle)**, shared across all four stages so scores are directly comparable. Because `StratifiedKFold` 's split depends only on the target vector and seed, the July folds reproduce the Feb v1 folds **byte-identically**, letting the reused Feb anchor sit on the same ladder. The leaderboard is closed (late submission returns 403), so all scores here are **CV-only**; Feb's best real submission (v3, on a non-canonical multi-seed scheme) posted Public LB AUC 0.95332 as a loose sanity anchor.

Models & Architecture

Purpose of Architecture. A binary heart-disease classifier that maximizes ROC-AUC. **Architecture Type** is a **gradient-boosted decision-tree ensemble** — LightGBM as the primary learner, with XGBoost and CatBoost pool members, combined by a weighted blend; the champion is a **37-member mega-blend** discovered at tree-search node #43.

Each member takes the same **Input Format** — a numeric feature matrix (no categorical encoding needed; all columns are already numeric) — of **Input Dimension 13 features** for the champion LGB family (33 for the XGB/CAT members); a 1-D vector with no spatial/sequence structure.

Architecture Description. Each GBDT member is an additive ensemble of decision trees grown by histogram-based, leaf-wise gradient boosting (representative stage-2 LGB: ~646 boosting iterations; the Optuna-tuned champion LGB uses `num_leaves` = 102 with a shallow `max_depth` = 4 and strong regularization). The blend layer sits on top as a convex weighted average of members' out-of-fold predictions, with weights found by Dirichlet sampling plus coordinate-ascent refinement. **Model Complexity** is therefore best expressed as trees × leaves rather than a dense parameter count: the champion blends 37 pool members, though the weight search kept only **8 non-zero members** — **all LGB-family variants** of the tuned base-13 LGB, so in practice the mega-blend is an intra-LGB-family variance reduction.

Training procedures

Training proceeds as a **four-stage ladder** on the shared 5-fold split, each stage measured on the same OOF scale:

Stage	Configuration	OOF AUC
1	Feb v1 LGB+XGB anchor (34 features, fold-identical, reused not retrained)	0.95498
2	skill pipeline: 33-feature LGB+XGB+CAT blend (weights 0.15 / 0.25 / 0.60)	0.955197
3	linear iteration: base-13 prior check + <code>is_unbalance</code> ablation + Optuna 40-trial tune + seed-bag, 5-way blend	0.955510
4	tree-search <code>harness_v3</code> (60 nodes, node #43 = 37-member mega-blend)	0.955529

Two experience-library **priors were re-verified by controlled same-folds comparison** in stage 3: (A) "explicit product/threshold columns add nothing for GBDT" (s3e9) was **confirmed** — the 13 raw features (0.955238) beat the full 33-feature LGB (0.955009) by +0.000229, the opposite verdict to s6e1 but the same as s5e10; (B) "class-imbalance weighting hurts a ranking metric" (s3e3/s4e1) was **confirmed a third time** — `is_unbalance` = True (0.954993) lost to the default (0.955009) by −0.000016. The **Loss Function** is binary log-loss (LightGBM `binary` objective) with model selection on AUC; the default **unweighted** objective was kept. The **Optimization Algorithm** is gradient boosting (histogram, leaf-wise) — **not** SGD/ADAM — with blend weights optimized by Dirichlet search + coordinate-ascent.

The **Learning Rate** is the boosting shrinkage (tuned champion LGB `learning_rate` ≈ 0.0171); there is **no Learning Rate Scheduler** (N/A — GBDT convergence is governed by CV early-stopping, not a schedule) and **no Batch Size** (N/A — full-dataset histogram boosting, not mini-batched).

Training Duration for the stage-4 tree search was **~7,901 s of wall-clock over 60 evaluated nodes** (0 failed, 0 dedup rejections; the search stopped on the 60-node hard budget cap); the stage-3 Optuna tune was 40 trials / 515.1 s on a fold-0 proxy. **Training Memory** was not explicitly capped — a $630k \times 33$ float matrix is comfortable on the single arm64 CPU machine; peak was not separately recorded.

Transfer Learning is N/A (no pretrained weights); its analogue here is **cross-season experience injection** — distilled priors from s3e3/s3e9/s5e10/s6e1 carried in and each re-verified on this data. **Data Augmentation** is N/A (tabular); the analogues are feature engineering and seed-bagging.

Reproducibility Standards are strict: fixed seeds (fold seed = 42; Optuna seed and seed-bag seed = 2024 logged), LightGBM determinism flags (`deterministic=True`, `force_row_wise=True`, fixed `num_threads=16` across all models), pinned from the very first run (the hard-won s6e1 fix). This was a **native v3 run — no replay needed**; the stage-4 OOF-reproduction gate (`06_rebuild_tree_best.py`) retrained all 37 members and matched cached OOFs digit-for-digit ($\max|\Delta\text{OOF}| = 0$), recovering the blend AUC 0.955529 and the stored weights member-for-member.

Inference procedures

Decision Threshold is N/A — AUC is a ranking metric, so we submit raw positive-class probabilities and never threshold (a 0.5 cut would only be needed if hard labels were required). **Inference Duration** was not separately profiled; GBDT scoring of the 270,000 test rows is sub-second per member on CPU, and **Inference Memory** is negligible (tree inference is lightweight, not capped).

Evaluation & Benchmarking

Task for Performance Evaluation. Rank the test patients by heart-disease probability. **Performance Metrics.** ROC-AUC is the sole competition metric; our stage-4 champion reaches **CV OOF AUC 0.955529** (submission file `0.95553`), climbing $0.95498 \rightarrow 0.955197 \rightarrow 0.955510 \rightarrow 0.955529$ across the four stages (stage $1 \rightarrow 4 \approx +0.0575\%$). The whole ladder moves within a ~ 0.0006 AUC band — an honest "last-mile" competition where gains are real but small, and the mega-blend still extracted a final $+0.000019$ over the linear stage's best blend. The Feb v3 prior-season submission (non-canonical multi-seed CV) posted Public LB 0.95332 — reference only, as this run is CV-only.

Performance Benchmarking. The natural comparator is the **NVIDIA reproduce-agent**, which copies the strongest public kernel:

Agent	Approach	CV AUC	Note
NVIDIA	reproduces <code>kospintr/heart-xgb-lightgbm-catb-baseline-k-fold</code> (verbatim-ish, CPU)	0.9554	tie

Agent	Approach	CV AUC	Note
Our agent	from-scratch GBDT pool + tree-search v3	0.95553	nudges ahead

The result is effectively a **dead heat**, with our from-scratch agent fractionally ahead of the reproduced public kernel — consistent with the newer-season (S5–S6) pattern where the reproduce-vs-originate gap collapses, in contrast to the small public-kernel edge seen on older S3-era problems. Both numbers are leak-free.

Final-Architecture Re-Run (v5 pipeline, Stage 0.5 in-lane) — 2026-08

All 20 baseline competitions were re-run in an isolated root (`~/benchruns/myagent-rerun`) under one frozen architecture (`docs/rerun_manifest.json`): every lane runs the **full v5 pipeline including the Stage 0.5 problem dossier** (`dossier.json` written in-lane before modeling), holds no credentials, and is checked by a per-lane transcript audit. Leaderboard scores for the re-run were produced by the separate credentialed operator scoring step and are recorded in `benchmark_results/rerun_three_way.csv` ; the figures below revise nothing in the sections above.

- **Lane:** ran and was scored.
- re-run local CV (lane `STATUS.md`): **OOF ROC-AUC 0.955418** — 27-member blend (6 with non-zero weight), tree-search node #43, 5-fold StratifiedKFold (shuffle, seed 42).
- Public / private leaderboard for this re-run (`benchmark_results/rerun_three_way.csv`): **Public 0.95354 / Private 0.95505**; `winner_priv` = **aide** (AIDE 0.95516, NVIDIA 0.95508).